Blue/Green Deployments with Elastic Beanstalk
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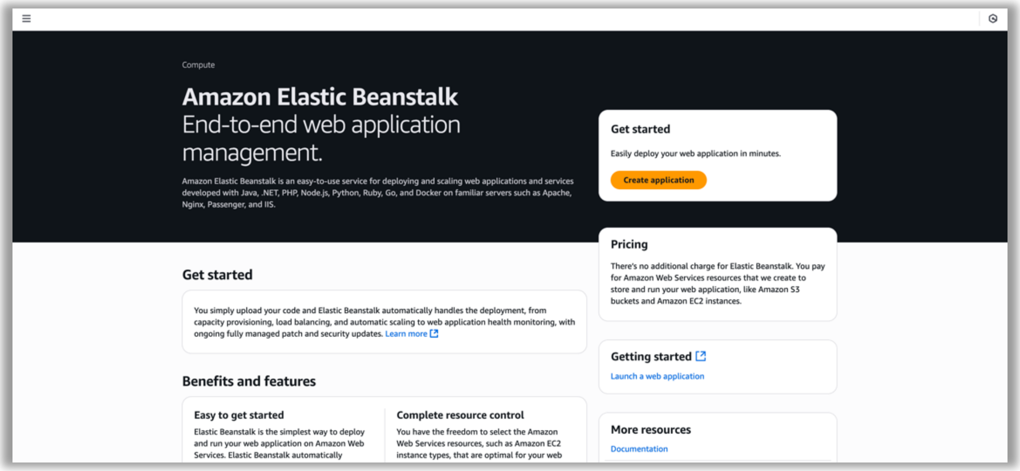
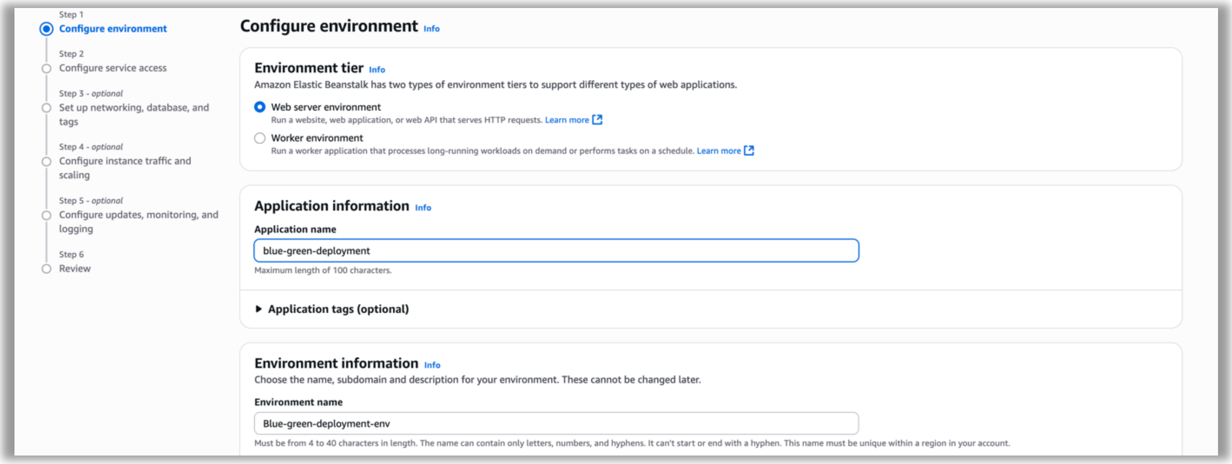
Introduction
This project demonstrates the implementation of a Blue/Green deployment strategy using AWS Elastic Beanstalk. The goal was to avoid disruptions to live applications while testing new changes. The process involved creating and cloning Elastic Beanstalk environments, uploading application versions, and performing a CNAME swap to transition traffic.

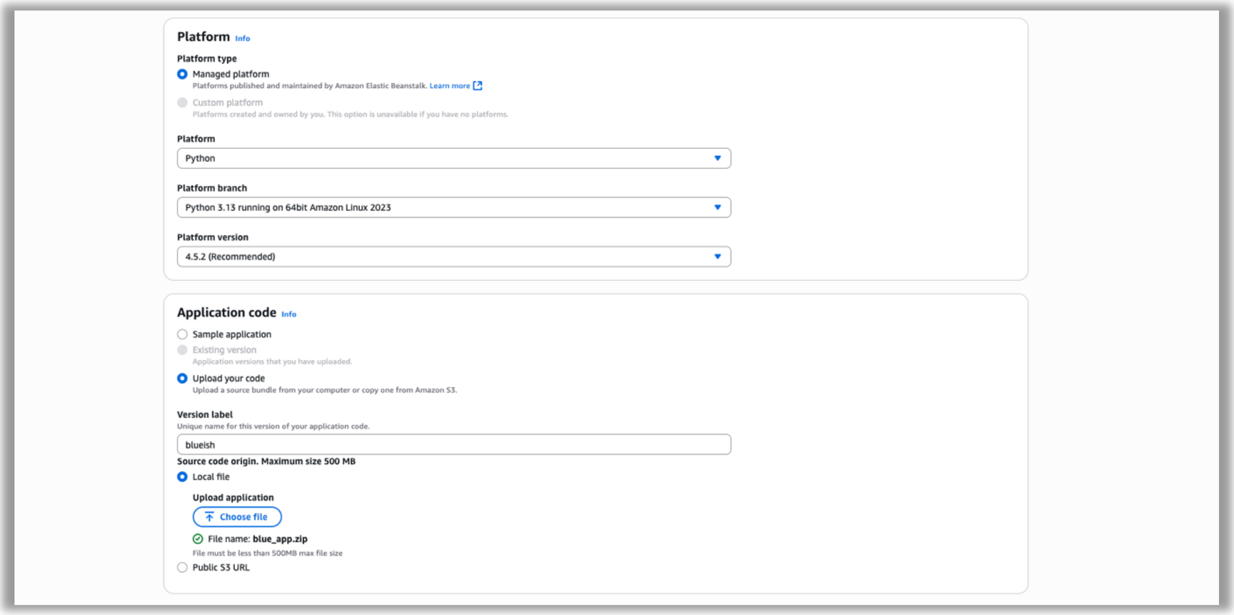
Steps Completed
1. Log In to AWS Console
- Accessed the AWS Management Console using provided credentials.
- Ensured the selected region was us-east-1.
- Opened the Elastic Beanstalk service.



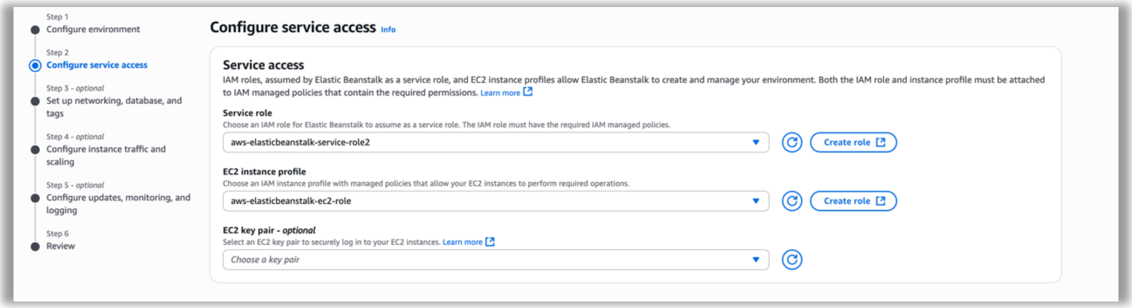
2. Create Elastic Beanstalk Environment
- Clicked Create application.
- Entered blue-green-deployment as the Application name.
- Chose a unique domain (e.g., blue1234) and verified availability.
- Set environment name to Blue-Deployment.

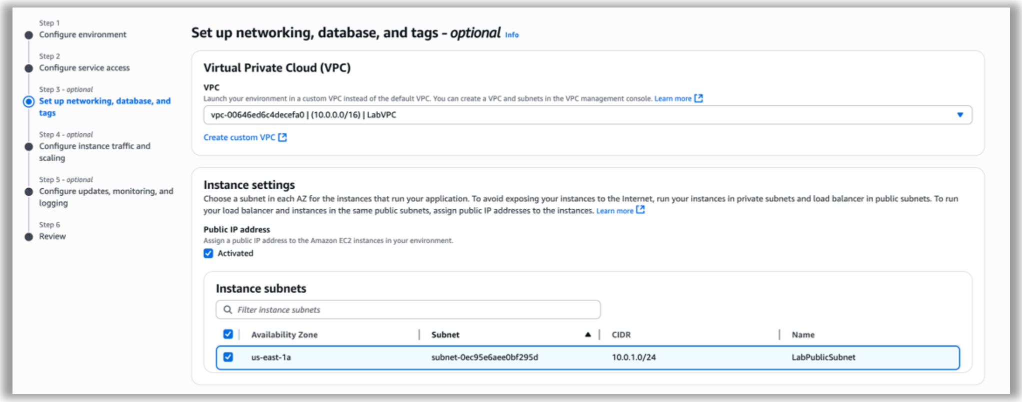


- Selected the Python platform.
- Uploaded the blue_app.zip file.
- Assigned the version label blueish.

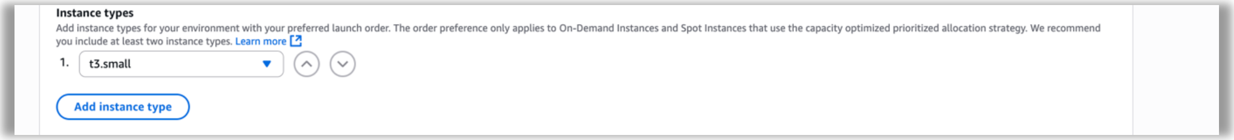


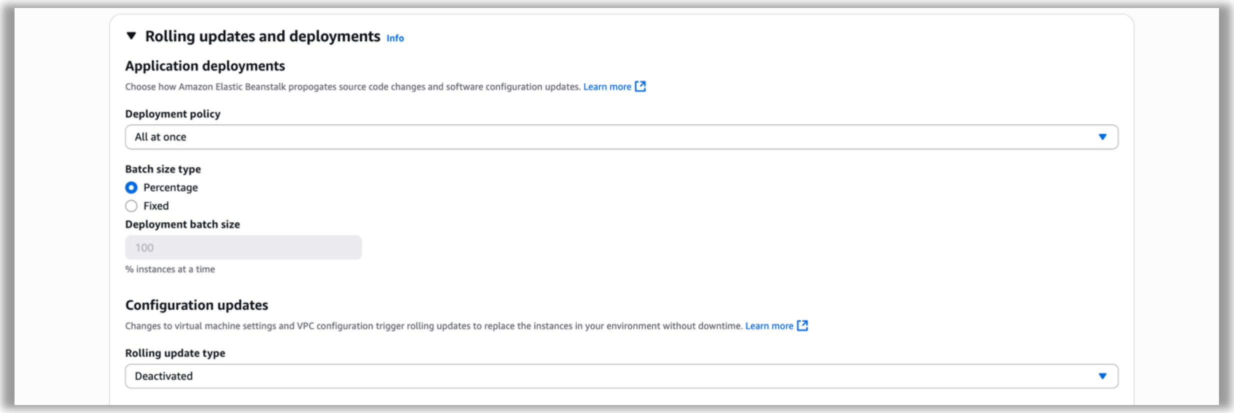
- Chose Custom configuration.
- Selected the existing service role aws-elasticbeanstalk-ec2-role.
- Chose the custom VPC LabVPC, activated Public IP, and selected subnets.

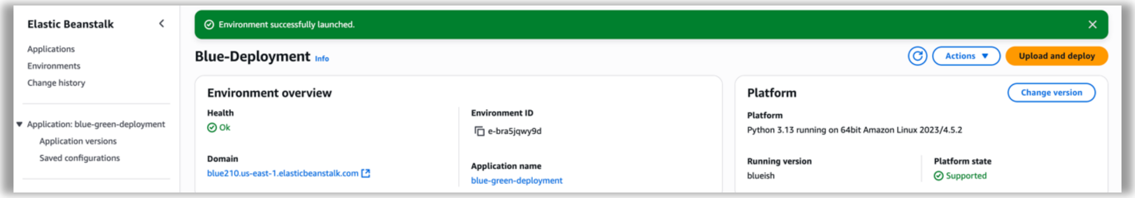




- Removed the t3.micro instance type.
- Skipped logging and monitoring configuration.
- Submitted the deployment and verified the application was correctly deployed via the domain link.

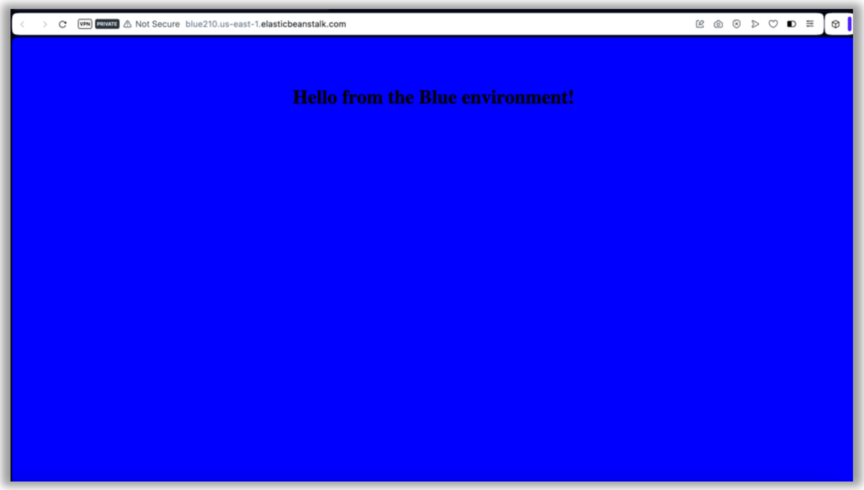


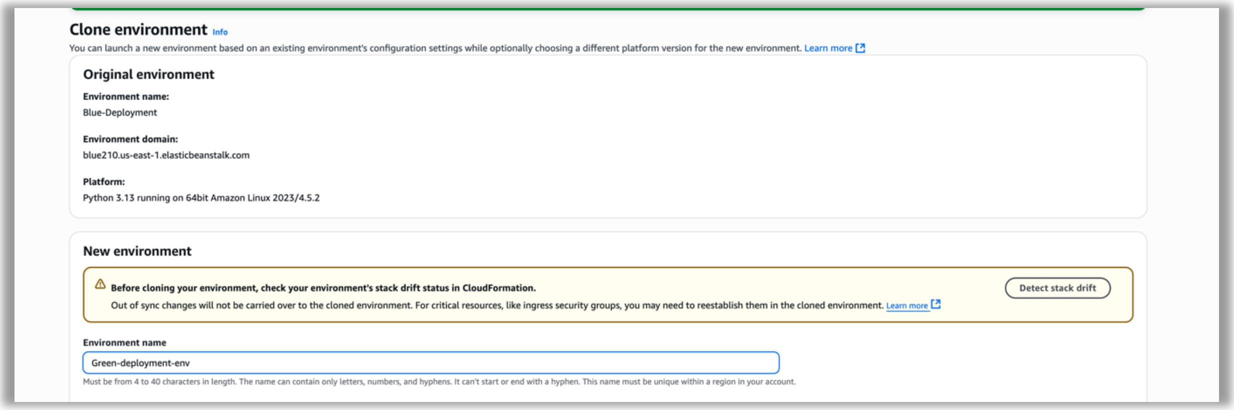


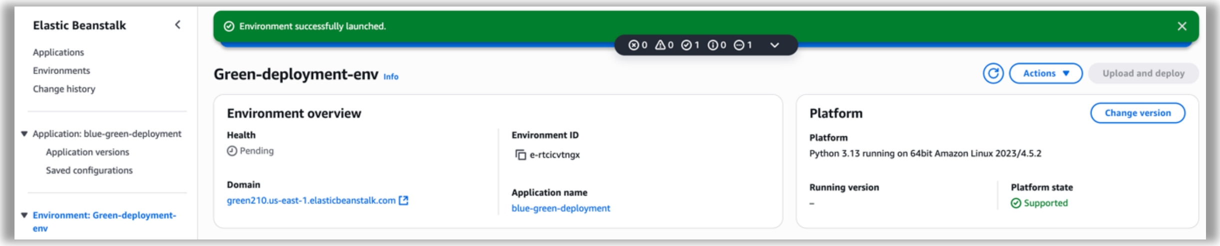


3. Clone Elastic Beanstalk Environment

- After confirming Blue-Deployment was active, selected Actions > Clone Environment.
- Renamed environment to Green-Deployment.
- Entered a unique domain (e.g., green5678) and verified availability.
- Updated the environment description.
- Selected the same service role (aws-elasticbeanstalk-ec2-role).
- Initiated the clone process.

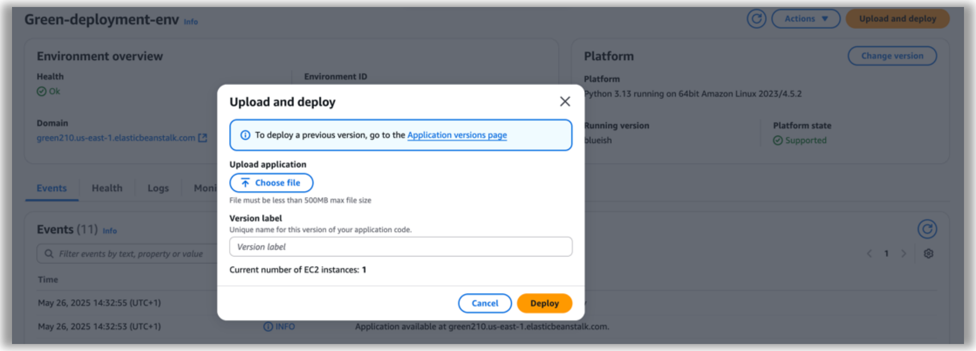


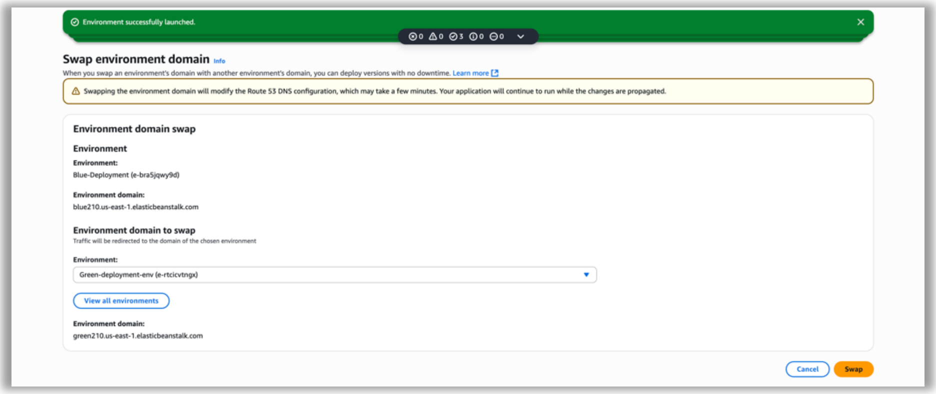


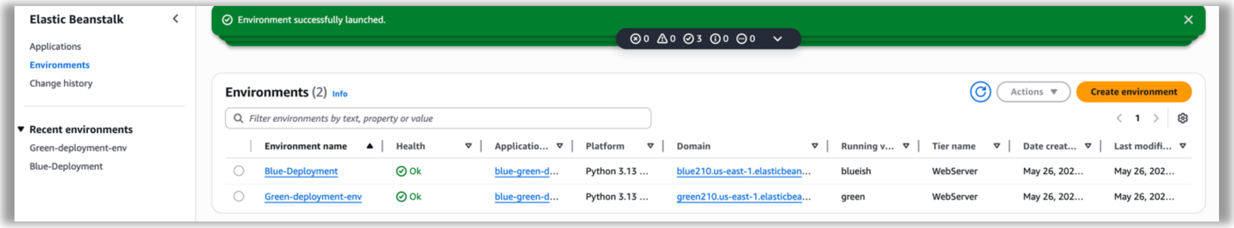


4. Deploy Green Application

- Once Green-Deployment was active, clicked Upload and deploy.
- Uploaded the green_app.zip file.
- Labeled the version Green.
- Deployed the green application version.
- Verified deployment via the domain link.







5. Swap CNAMEs

- Navigated to the Elastic Beanstalk environments dashboard.
- Selected Blue-Deployment and clicked **Actions > Swap environment domain**.
- Chose Green-Deployment and confirmed the swap.
- Verified that the Blue-Deployment domain now pointed to the green application.

Conclusion

The blue/green deployment was successfully implemented using AWS Elastic Beanstalk. The environment was duplicated, modified, and tested independently before traffic was redirected using a CNAME swap. This approach ensured zero-downtime deployment and a rollback-safe update mechanism.